

Abstract algebra – 1

Finite groups: counting

Homework

13 сентября 2026 года

Theoretical minimum. A group is a set with an associative operation, a neutral element and inverses; the order $\text{ord}(x)$ of an element is the least $k > 0$ with $x^k = e$; the index $[G : H]$ is the number of cosets xH . Lagrange's theorem says that the cosets partition the group, so $|G| = [G : H] |H|$, whence $\text{ord}(x)$ divides $|G|$ and $x^{|G|} = e$. Cauchy's theorem says that if a prime p divides $|G|$ then G has an element of order p ; for $p = 2$ it follows from the count $\#\{x : x^2 = e\} \equiv |G| \pmod{2}$, obtained by pairing x with x^{-1} . The class equation $|G| = |Z(G)| + \sum [G : C(x_i)]$ shows that a group of order p^k has a non-trivial centre; if $G/Z(G)$ is cyclic then G is abelian. Every finite abelian group is a direct sum of cyclic groups $\mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$ with $d_1 \mid \cdots \mid d_k$, and d_k is the exponent; a group with $x^2 = e$ for all x is isomorphic to $(\mathbb{Z}/2)^k$.

1. Let r, s, t be pairwise coprime positive integers and let a, b be elements of a commutative group with $a^r = b^s = (ab)^t = e$. Prove that $a = b = e$. Does the same conclusion hold in an arbitrary non-commutative group? (IMC)
2. Prove that a group G in which exactly two elements other than the identity commute with each other is isomorphic to $\mathbb{Z}/3\mathbb{Z}$ or to S_3 . (Румыния, 1996)
3. Let G be a finite group of order n , let $H = \{x \in G : x^2 = e\}$ and $p = |H|$. Prove that (a) $|H \cap xH| \geq 2p - n$ for every $x \in G$; (b) if $p > \frac{3n}{4}$ then G is commutative; (c) if $\frac{n}{2} < p \leq \frac{3n}{4}$ then G is not commutative. (Румыния, 2010)

4. For a finite group G let $n(G)$ be the number of its elements and $s(G)$ the number of its subgroups. Decide whether each of the following is true: (a) for every $a > 0$ there is a finite group with $n(G)/s(G) < a$; (b) for every $a > 0$ there is a finite group with $n(G)/s(G) > a$. (Румыния, 1999)
5. Let G be a group with m elements and H a proper subgroup with n elements. Put $H^x = \{xhx^{-1} : h \in H\}$ and assume $H^x \cap H = \{e\}$ for every $x \in G \setminus H$. Prove that $H^x = H^y$ if and only if $x^{-1}y \in H$, and find $|\bigcup_{x \in G} H^x|$ as a function of m and n . (Румыния, 2005)
6. Let $*$ be a commutative and associative binary operation on a set S such that for all $x, y \in S$ there exists $z \in S$ with $x * z = y$. Show that $a * c = b * c$ implies $a = b$. (Putnam 2012, A2)
7. Prove that if $G/Z(G)$ is cyclic then G is abelian, and deduce that every group of order p^2 is abelian. Then show that for every prime p there is a non-abelian group of order p^3 , so the statement is sharp.
8. Let $\pi: G \rightarrow Q$ be a surjective homomorphism of finite groups. Prove that there exists a subgroup $H \leq G$ such that $\pi(H) = Q$ and the set of primes dividing $|H|$ coincides with the set of primes dividing $|Q|$. (IMC 2025, shortlist)
9. Let n be a squarefree positive integer and G a group of order n . Let d be the largest order of an element of G and N the number of elements of order d . Prove that (a) N is divisible by n/d ; (b) $N^2 \geq \varphi(n)$. (Румыния, 2026)
10. Let G be a finite group with the property that for every automorphism f of G there exists $m \in \mathbb{N}$ with $f(x) = x^m$ for all $x \in G$. Prove that G is commutative. (Румыния, 2017)